SmartSDR™ for Windows v3.9.18 Release Notes

May 28, 2025

SmartSDR™ v3.9.18 is a general release for all FLEX-8000 and FLEX-6000 series software-defined radios, and the Maestro™. This version supersedes all previous versions of SmartSDR v3.x.

This document covers changes to client software (SmartSDR for Windows, SmartSDR for the "M" radios, and Maestro). Changes to clients or utilities produced by 3rd parties are not addressed. Please consult the SmartSDR Software User Guide v3 for details on using features and an optimal operating experience.

Before upgrading, please read all sections marked as IMPORTANT for upgrade prerequisites and other important information. Failure to follow these instructions may create problems requiring FlexRadio support or service staff assistance.

Note: Recent versions of SmartSDR v3 address several critical software defects identified in previous versions of SmartSDR. After upgrading to SmartSDR v3.9.18, downgrading to previous versions of SmartSDR v3 will result in the reoccurrence of critical issues resolved in this software release.

Important Notice: After upgrading the v3.9.18 firmware in all FLEX-8000s and FLEX-6000s a hardware power cycle, which includes shutting down the radio, cycling the DC power, and restarting the radio is required to ensure the firmware update installs successfully.

New Features in SmartSDR v3.9.18

Introducing SmartSignal™ (Beta): FlexRadio is pleased to announce the beta release of SmartSignal, our new automatic adaptive pre-distortion (APD) feature designed to significantly reduce intermodulation products at the transmitter output. SmartSignal for the FLEX-8000 actively analyzes and compensates for amplifier non-linearity in real time, improving spectral cleanliness and transmitter performance.

Important: Please refer to the Caveats and Known Issues section below for information regarding the SmartSignal Beta limitations in SmartSDR v3.9.18.

To enable SmartSignal, first check to see if the APD status shows AVAIL, indicating the APD meets the requirements for use. Next, simply enable SmartSignal in the SmartSDR software by clicking/enabling the APD button. The APD status will change to CAL indicating SmartSignal is ready for calibration. The last step is calibrating or training SmartSignal for optimal signal quality by keying the mic while speaking when in SSB

mode. When training is complete, usually taking less than a second you will see the APD status change to ACTIVE. If at any time you see the status change to CAL, simply transmitting in SSB mode will reactivate SmartSignal. For additional information, please refer to the SmartSignal FAQ available from the HelpDesk: SmartSignal™ (automatic adaptive pre-distortion) FAQ.

As a **beta feature release**, this version is not intended to be perfect, and it is being released to gather feedback and generate data for future improvements to the underlying algorithm. We encourage you to share this information with users and support teams via the HelpDesk to help manage expectations and provide clarity on the current state of SmartSignal. We look forward to your feedback and suggestions for further development.

Analog Meter Display: SmartSDR v3.9.18 now provides a combo analog meter option in addition to the standard bar meters. The feature is enabled in the right-hand side applet bar by clicking on the ANLG tab located at the top of the applet bar. You can select either the normal or peak reading version of the meter. When receiving, the meter displays the signal level in S units. The available transmit meters are Power, SWR, Level (audio input), and Compression.

USB Cable Serial Pass-Through Feature: With SmartSDR v3.9.18 and above, FlexRadio introduces Serial Pass-Through, a USB Cable feature that allows you to connect and control RS-232 serial devices, such as antenna rotators and amplifiers, directly *through* your radio. This feature works locally and over the internet using SmartLink. For additional information and instructions on how to use and configure the Serial Pass-Through feature, please refer to the HelpDesk article SmartSDR Serial Pass-Through Feature.

The Automatic Profile Save Feature can be disabled: There are certain circumstances where the profile (microphone and TX) auto-save feature may not be ideally suited for your operating workflow. In these instances, SmartSDR now provides the ability to disable the profile auto-save action. Enabling and disabling the profile auto-save capability is located in the top option bar in the SmartSDR Profile Manager. When disabled, if a change is made to a Microphone or TX Profile parameter, a small floppy disk icon is shown to the right of the Microphone and TX profileselection drop-down box on the Phone/CW (p/cw) sidebar indicating a parameter has been changed. Clicking on the icon will save the changed profile.

Two-Tone Signal Generator: A two-tone signal generator has been added to SmartSDR. To utilize the two-tone generator, you must select between the single and two-tone signal generation options in the Transmit settings. To activate the two-tone generator, simply click or press the Tune button.

TGXL Integration into SmartSDR for Windows and Maestro: As a companion to the PGXL integration, you can now control the basic functions of the TGXL directly from SmartSDR. Please note that to utilize this feature, the TGXL must be running version 1.2.17 or greater of the TGXL Utility. The TGXL Utility can be downloaded from the 4O3A website.

Bug Fix Highlights in SmartSDR v3.9.18

Please refer to the SmartSDR v3 Changelog for a complete listing of changes.

[SMART-10697] 8000 Radios: Receive Audio goes silent after transmitting with NR and ANF enabled

[SMART-10633] SmartSDR CAT frequency no longer gets out of sync following direct Frequency input from a SmartSDR client

[SMART-11048] Improvements to SmartLink connection security

[SMART-10698] SmartSDR will now display 'GNSS' vs 'GPS' on the status indicator of the main view

[SMART-10785] Enabled 10MHz output when GPSDO is installed on the 8000 Series Radios

[SMART-10998] 8000 Series Radios with GPSDO now show the correct GPS status

[SMART-10757] Changed wording in Frequency Offset menu to show 'GNSS is installed' for 8000 radios with the stock option and 'GPS is installed' for radios with the optional GPSDO upgrade.

[SMART-10463] Frequency Offset is now available for external 10 MHz and TXCO. Values are saved uniquely for each source; Frequency Offset is not offered for GPSDO or GNSS.

[SMART-10969] Fixed an issue with panadapter resizing that incorrectly reported a UHE: Out of Memory errors

[SMART-10437] Fixed an issue where DAX channels intermittently get out-of-sync following a sequence of events and Band changes. Related to SMART-10640

[SMART-10672] Improvements to network disconnect handling on both radio and clients

[SMART-10734] 8000 radios can now use SmartLink when the radio is assigned a Static IP

[SMART-9562] Fixed an issue where the subnet mask value was incorrectly displayed after entering a static IP address

[SMART-9226] Various improvements to meter ballistics for existing digital meters.

[SMART-10903] Fixed an issue in PROC that made compression less effective on 8000 Series Radios

[SMART-11001] Moved AGC to just before modulation in the audio signal chain

[SMART-10649] Fixed an issue with 8000 Series Radios that caused low RF power out levels on XVTR B

[SMART-10074] Fixed a memory leak caused by loading a specific profile

[SMART-9650] Changed Maestro and M Model error handling forcing a reboot if a fatal error has occurred

[SMART-10740] Improved M Model and Maestro notifications and handling of PIC software updates

[SMART-10113] Fixed a UHE and incorrect Warning Messages when changing the network with a Maestro connected to a radio

Important Release Details

There are minimum versions of SmartSDR needed for certain FLEX-6000, FLEX-8000, and Maestro models due to hardware revision levels. The table below indicates the minimum required version of SmartSDR based on hardware ship dates.

Using the latest version of SmartSDR will prevent any issues you may encounter by running firmware below the minimum version for your radio.

Product	Hardware Ship Date	Minimum Version
Maestro C, FLEX-8400M FLEX-8600M	17-February-2025	3.8.23
FLEX-8400M, FLEX-8600M	2-January-2025	3.8.21
FLEX-8400, FLEX-8600	9-October-2024	3.8.19
FLEX-6400, FLEX-6400M, FLEX-6600, FLEX-6600M	24-March-2023	3.4.23

What to do if the SmartSDR software update does not finish after 10 minutes? If ten (10) minutes have elapsed when the upgrade indicator is no longer moving and the radio has not rebooted to complete the upgrade, power off the radio by pressing the power button. If the power button is unresponsive, monmentiarly remove DC power from the radio to shut it down. The radio should restart, and the firmware update should have been completed successfully.

What to do if the FLEX-8000/6000 fails to boot after upgrading to SmartSDR 3.9.18: If the radio fails to boot after performing a firmware upgrade, please open a HelpDesk support ticket to get the issue resolved.

Upgrading an M-model Radio Front Panel: If you are upgrading an M-model radio using SmartSDR for Windows, the upgrade will not be applied to the software for the Front Panel (display). Use the information below to upgrade the front panel.

- ***The radio must first be connected to your network and can access the Internet to download the necessary software to update the front panel.***
- After the radio firmware updates from your PC, shut down SmartSDR for Windows and connect to your radio using the M-model Front Panel. The Version Select screen should display SmartSDR v3.9.18 showing a download icon. Selecting that version will start the final portion of the upgrade process.

For a detailed description of the SmartSDR for Windows software upgrade process, please refer to the SmartSDR Software Users Guide v3

Upgrading the radio firmware over a SmartLink connection is not supported. Since Internet connections can be inherently unreliable and it is paramount that the firmware upgrade process proceeds uninterrupted to prevent the FLEX-8000/6000 from becoming inoperable, upgrading the radio firmware is only supported using a LAN or network-connected SmartSDR client.

Best Practices for Installing / Uninstalling SmartSDR

Best Practices for Installing a New Version of SmartSDR for Windows

***Always* back up your Global, TX (transmit), and Microphone Profiles. For detailed instructions on exporting your profiles to a file on your PC, refer to the SmartSDR for Windows Software User's Guide**. It is recommended to export all profiles before upgrading to v3.9.18. Maintaining a good set of profile exports as backups is always a best practice. Moving back and forth between different versions of SmartSDR may result in data loss unless you have a backup (export) of your Profiles.

Power cycle the radio before installing a new version of SmartSDR for Windows. To ensure a seamless upgrade, it is recommended that you power cycle your radio *before* installing the SmartSDR for Windows software on your PC and updating the radio firmware.

Ensure your Windows operating system is up to date before installing SmartSDR for Windows. The proper operation of SmartSDR for Windows and its associated drivers relies on having an up-to-date and supported Windows operating system including the root security certificates. It is strongly recommended that you run Windows Update and install all mandatory and optional updates before installing SmartSDR for Windows.

Managing SmartSDR installed on other devices. If using Maestro or other PCs running SmartSDR for Windows, update all devices at the same time to ensure a consistent operating experience and to avoid radio firmware upgrade/downgrade delays.

"Cold Boot" the radio after upgrading the radio firmware. Using the power button, shut down the radio, then disconnect it from DC power for approximately 30 seconds. This procedure "*cold boots*" the radio and helps ensure proper operation. Once DC power is restored to the radio, wait 2 minutes to ensure all internal processors have booted up completely before booting your radio.

Always Perform a Factory Reset of your Radio when <u>Downgrading</u> the Radio Firmware: In general, downgrading to a previous version of SmartSDR is not recommended. The database in the FLEX-8000/6000 is NOT backward compatible. This means if you downgrade the firmware in your radio, you *must perform a Factory Reset* to ensure the internal database is consistent with the version of SmartSDR firmware that is running on the radio.

The procedure for performing a Factory Reset is described in the **<u>Best Practices for Installing a New Version of SmartSDR for Windows section</u>** below.

In addition to performing a Factory Reset on the radio, you mustn't import a database using a profile export that is greater than the version you are using with the FLEX-6000. In later versions of SmartSDR, the profile export file name contains the version of SmartSDR that was running when the export was saved to your PC for easy version identification.

How to Factory Reset a FLEX-8000/FLEX-6000

There may be situations where a factory reset of the radio is needed to ensure that the radio is working properly. If instructed to perform a Factory Reset of the radio, use the procedure below.

Perform a Comprehensive Cold Boot and Factory Reset of your radio.

NOTE: Before starting, shut down all programs that may interface with the radio such as loggers, digital mode programs, and NodeRed devices.

NOTE: If you have anything connected to the REM ON connector on the back of the radio, please remove it before running the reset procedure. The REM ON device can be reconnected once the reset procedure is finished.

Before you get started, please note the "wait times" in the steps listed below. These are important to ensure the radio resets properly.

- Power off the radio by pressing and releasing the power button. Allow it to completely power down before continuing by waiting for the power LED to turn off (or amber if an FLEX-6000 GPSDO is installed). If pressing and releasing the power button does not shut down the radio, **press and hold** the power button until the radio shuts down. If this does not work, turn off the DC power supply to shut down the radio.
- Once the radio is powered off, **wait for 2 minutes** to allow for all processors to shut down completely.
- Remove the power cable from the radio for **at least 30 seconds** and then reconnect it. It is important to remove the power cable from the radio and not just turn off the power supply.
- After reconnecting the DC power cable, turn the DC power ON and **wait for 2 minutes** to allow the internal PSoC processor to boot up completely before continuing.

For FLEX-6700 and FLEX-6500:

- Press and hold the **OK** button *while simultaneously pressing and releasing* the **Power button**.
- Release the **OK** button once the power LED turns white and allow the radio to continue booting normally. When the Power LED is solid green, the radio has completed the boot-up process.

For FLEX-6300, FLEX-6x00(M), FLEX-8x00(M):

- Press and hold the **Power button** for approximately 5 seconds until the Power button LED turns **white**. *After the power button turns white, release the **Power button*** and allow the radio to continue booting indicated by a flashing green power button LED. When the Power button LED is solid green, the radio has completed the boot-up process.

  Depending on several factors, it may take a few minutes for the radio to finish booting so please be patient and allow it to boot up completely.

Best Practices for installing a new version of SmartSDR for an "M" Model or Maestro

The following best practices are applicable only when installing a new version of SmartSDR on a Maestro.

Both the Maestro and "M' Model radios must have network access that allows connectivity to the Internet to download the new SmartSDR software.

Ensure your Maestro has a reliable power source. Make certain the supplied AC adapter is used to power the unit. This prevents the Maestro from losing power during an update.

Ensure your Maestro has a reliable network connection. The Maestro and radio firmware are upgraded entirely through the network connection. When upgrading a Maestro, a wired Ethernet connection is recommended; wired Ethernet connections are faster and more reliable than Wi-Fi.

Uninstalling Previous Versions of SmartSDR for Windows – Is It Necessary?

In general, the answer is no, but there are considerations when more than one version of SmartSDR is maintained on your system:

Currently, every version of SmartSDR for Windows is installed independently of each other permitting the use of previous versions and supporting convenient version switching of both software and radio firmware as long as there are no software or database dependencies that prevent backward compatibility.

SmartSDR for Windows Shared Components:
The SmartSDR software employs shared components used by all versions of the software. When previous versions of SmartSDR for Windows are uninstalled it may result in the removal of one or more of these shared components, which may make newer versions inoperable.

Therefore, if you desire to uninstall an older version of SmartSDR, we highly recommend uninstalling all previous versions of SmartSDR before installing a new version of SmartSDR for Windows.

Important: Removing the DAX and FlexVSP drivers <u>is not recommended</u> unless explicitly instructed in the Release Notes or by FlexRadio Support.

Caveats and Known Issues

SmartSignal Beta

Important: The following caveats apply to the beta version of SmartSignal.

• **APD Generation:** The Automatic Adaptive Pre-Distortion is generated from the transmitting slice. In a multi-Slice scenario, this can become more complicated but typically follows the Local PTT TX slice.
• **Internal Sampler Requirement:** Currently, SmartSignal can only be utilized alongside the internal sampler in the FLEX-8000.
• **Works Best with Default RF Gain Settings:** SmartSignal operates optimally when the RF gain settings are kept at their default values.
• **Best Performance with Voice Signals:** The feature achieves the highest success rate when employed with voice transmissions.
• **Challenges with 6M Band:** Generating an equalizer on the 6-meter band can be more complicated and less reliable.
• **Challenges with AM Signals:** Similar to the 6M band, creating an equalizer with AM signals is also more challenging.
• **Limited Performance in multiFLEX Scenarios:** The feature exhibits quirks in multiFLEX scenarios and may not deliver the desired results.
• **Occasional APD Stuck in "Calibrating Mode"**: There are instances where the APD may become stuck in "calibrating mode." This can happen if it cannot identify a better solution than a previously stored one. If this occurs, rebooting the radio may be necessary to resolve the issue and enable the APD to reattempt the calibration.

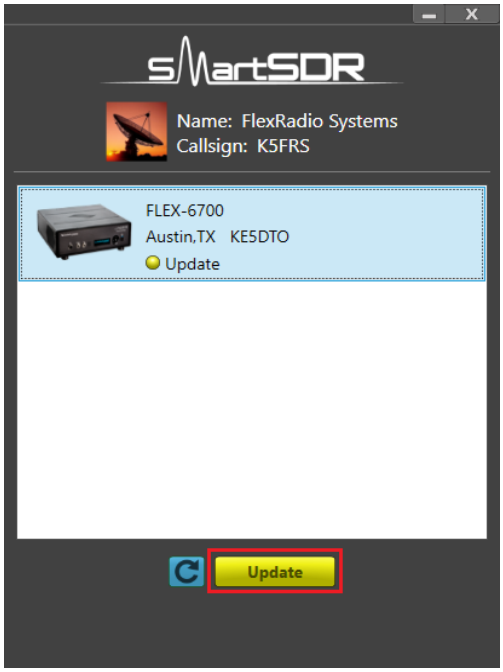
Reverting to previous versions of SmartSDR v3 is not recommended. However, if you revert from any version of SmartSDR v3 to previous versions you must perform a factory reset of your radio. Database incompatibilities may result in operational anomalies when reverting to previous versions of SmartSDR. It is recommended that a factory reset be performed after downgrading to ensure the database schema is 100% compatible with the version being used.

[SMART-11274] SmartControl intermittently loads incorrectly when anything other than a single Panadapter is in use. Workaround this issue by closing a Panadapter (or opening one) to have only one active panadapter, then make any adjustments you need in the main SmartControl window area.

Once the radio has been factory reset, **DO NOT** import any profiles created using a version of SmartSDR greater than the one installed on the radio. We recommend deleting any profiles exported using SmartSDR v3.2.31 to ensure they are not utilized as profile backups. Importing profiles created with earlier versions of SmartSDR than the installed version is acceptable.

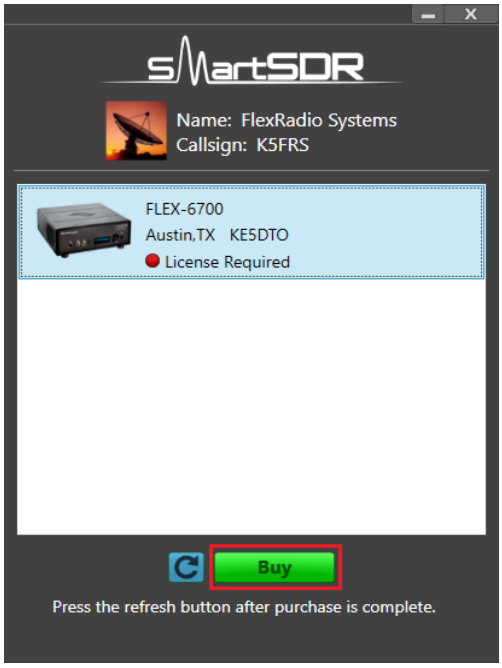
Purchasing a SmartSDR Software License for your FLEX-6000 using SmartSDR for Windows

Obtaining a SmartSDR software license using SmartSDR for Windows is easy and convenient. Download the latest software installer for SmartSDR and run it to install the software on your PC. Using the new software, update the radio with the new firmware. Start SmartSDR, select the radio in the radio chooser window, and click the yellow Update button.



Once the update completes, if the radio you are upgrading requires a license, the radio status will show "License Required" and you will be presented with a Buy button. Clocking on the Buy button launches a web browser, which will walk you through the steps to purchase the license.

Note: New radios purchased include a SmartSDR license for the current major version and will not display the "License Required" indicator.

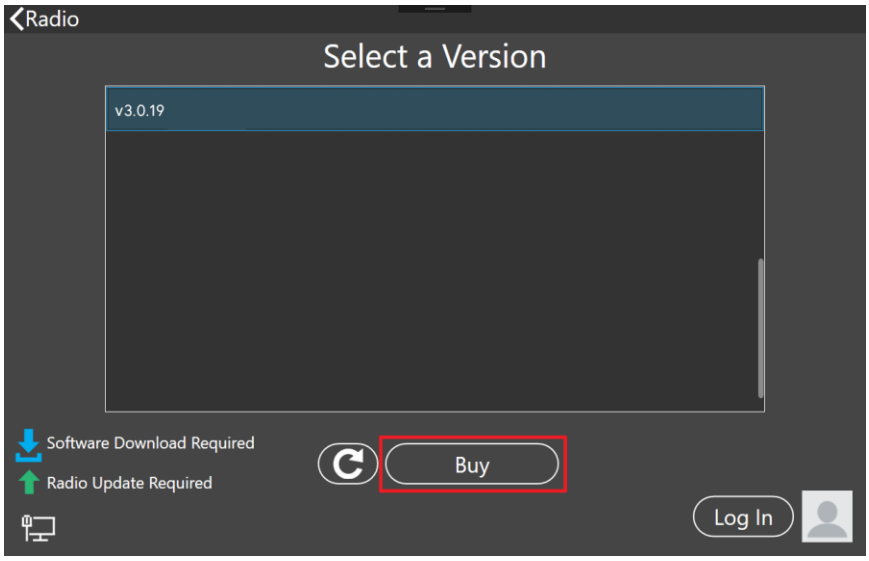


Once you have purchased a SmartSDR license for your radio, click the blue circular refresh button in the radio chooser window to download the license information for the radio. It may take up to a minute after the purchase before the radio acquires its license and cycling the power on the radio may be required.

Note: The radio (not just the computer) will need internet access to obtain the newly acquired license. Please ensure that the radio is plugged into a network with internet access.

Purchasing a SmartSDR Software License for your FLEX-6000 using an "M" Model FLEX-6000 or a Maestro

Obtaining a SmartSDR software license using an M-model radio or a Maestro is easy and convenient. After downloading and installing the latest SmartSDR software version. At the Select a Version screen, choose the version of the software just installed. If you need to purchase a license to run the software, the Buy button will appear as shown below.



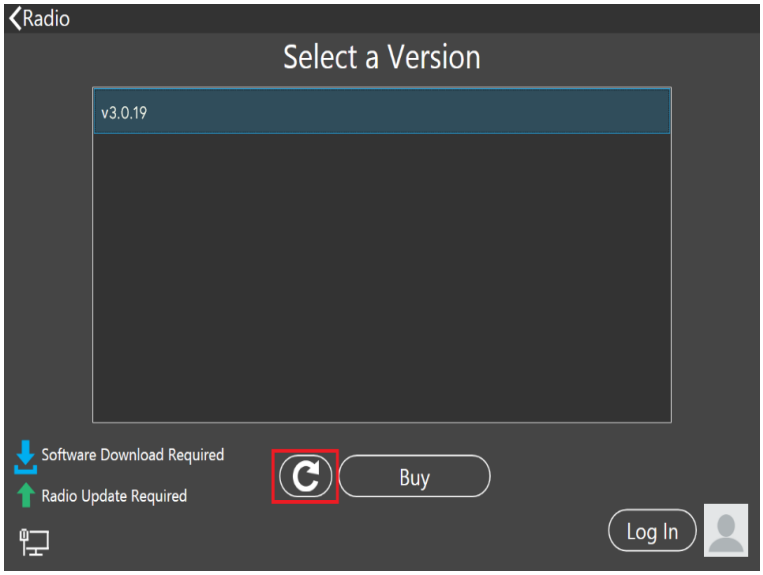
If you see a Run button, tap on the Radio link at the top left of the screen to return to the Select Radio to upgrade the radio's firmware.

Tapping the Buy button will display a radio-specific QR code.

- Open the camera app on your phone or tablet.
- Point the camera at the QR code and center it in the frame.
- Wait for a notification or banner to appear.
- Tap the notification or banner to open the link associated with the QR code.
- Follow the instructions to purchase a SmartSDR license upgrade.

When the license purchase is complete, tap the Version link at the top left corner of the screen to go back to the Select a Version screen.

At the Select a Version screen, tap the refresh button (circular icon) to download the license information for the radio. It may take up to a minute after the purchase before the radio acquires its license and cycling the power on the radio may be required.



After the license is updated, the refresh button will disappear and the Buy button will change to a Run button. Tap the Run button to connect to your radio.

SmartSDR Documentation Available from www.flexradio.com

The following documentation and how-to guides for SmartSDR v3 are available as a convenient download from the FlexRadio website.

SmartSDR for Windows Software User's Guide
FLEX-6400M and FLEX-6600M User's Guide
SmartSDR CAT User Guide
FLEX-6000 Signature Series Hardware Reference Manual
FLEX-6400 and FLEX-6600 Hardware Reference Manual
FLEX-6000 Signature Series Quick Start Guide
SmartLink for SmartSDR Quick Start Guide
USB Cable Interface Guide
SmartSDR v3 with multiFLEX Installation Video
SmartSDR v3 Changelog

Obtaining Technical Support

If you encounter any issues installing or operating SmartSDR for Windows with your FlexRadio Signature Series software defined radio, please use our online Community to search for information about SmartSDR and the FLEX-8000/FLEX-6000. Refer to the Community Message Board for assistance using the Community.

If you are unable to find an existing answer to your issue via the Community, please contact FlexRadio Technical Support by opening a HelpDesk support ticket online. Refer to the HelpDesk article How to Submit a Request for Technical Support for details on how to submit a HelpDesk support ticket,

Hours of Operation: Our Technical Support engineers are available Monday through Friday from 7:00 AM to 4:00 PM Central Time. If you open a HelpDesk ticket after business hours, on a holiday, or weekend, we will respond to your request for assistance during regular business hours in the order your HelpDesk ticket was received.

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